NFS :

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Configuring NFS Server and Client on RHEL

What is NFS?

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- The ***Network File System (NFS)*** is a network file system commonly used by UNIX/LINUX systems and network-attached storage devices to allow multiple UNIX/LINUX clients to share access to files over the network.
- It provides access to shared directories or files from client systems.
- Network File System (NFS), allows remote hosts to mount file systems over a network and interact with those file systems as though they are mounted locally.
- This enables system administrators to consolidate resources on centralised servers on the network.

NFS Versions :

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NFSv2:

- NFS version 2 (NFSv2) is older and widely supported.
- NFSv2 is not supported from RHEL 7 onwards
- It can use both TCP and UDP protocol over an IP network(port 2049).
- But it uses UDP running over an IP network to provide a stateless network connection between the client and server.

- UDP is stateless, if the server goes down unexpectedly, UDP clients continue to saturate the network with requests for the server.
- when a frame is lost with UDP, the entire RPC request must be re-transmitted.
- By default does not have any support for user based authentication
- Used to use ACL to manage the host based access

NFSv3:

- NFS version 3 (NFSv3) supports safe asynchronous writes and is more robust at error handling than NFSv2
- it also supports 64-bit file sizes and offsets, allowing clients to access more than 2GB of file data.
- It can use both TCP and UDP protocol over an IP network(port 2049).
- But it uses UDP running over an IP network to provide a stateless network connection between the client and server.
- UDP is stateless, if the server goes down unexpectedly, UDP clients continue to saturate the network with requests for the server.
- when a frame is lost with UDP, the entire RPC request must be re-transmitted.
- By default does not have any support for user based authentication
- Used to use ACL to manage the host based access
- Access to files over the network used to get controlled through multiple services

NFSv4:

- NFS version 4 (NFSv4) works through firewalls and on the Internet
- no longer requires an rpcbind service

- supports ACLs, and utilises stateful operations.
- RHEL 6 supports NFSv2, NFSv3, and NFSv4 clients.
- When mounting a file system via NFS, RHEL uses NFSv4 by default, if the server supports it.
- It uses TCP protocol.
- With TCP, only the lost frame needs to be re-sent.
- For these reasons, TCP is the preferred protocol when connecting to an NFS server.
- The mounting and locking protocols have been incorporated into the NFSv4 protocol
- The server also listens on the well-known TCP port 2049.
- The rpc.mountd daemon is required on the NFS server to set up the exports.
- Can be integrated with LDAP+Kerberos based authentication mechanism for user authentication.

NFS Server Configuration :

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IP Address = 192.168.19.129

1. **Install nfs packages :**

```
[root@server1 ~]# yum install nfs-utils -y
```

2. **Activate the nfs service :**

```
[root@server1 ~]# systemctl enable --now nfs-server.service
```

3. **Allow nfs through firewalld :**

```
[root@server1 ~]# firewall-cmd --permanent --zone=public --add-service={nfs,mountd,rpc-bind}
success
```

4. **Configure Selinux mode :**

vim /etc/sysconfig/selinux

Replace enforcing to disabled

Save and exit

Restart the system

5. **Share the folder in /etc/exports file**

```
[root@server1 ~]# vim /etc/exports
```

Enter the sharing and permission details inside this file

```
/share1          *(rw,sync)
```

Save and exit

6. Restart the nfs service

```
[root@server1 ~]# systemctl enable --now nfs-server.service
```

7. To check the shared folder :

```
[root@server1 ~]# showmount -e
Export list for server1.example.com:
/share1 *
```

8. To see all nfs shares from a network location

```
raja@node1:~$ showmount -e 192.168.19.129
Export list for 192.168.19.129:
/share1 *
```

NFS Client Configuration :

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Client OS : Ubuntu :

Install nfs package :

```
apt install nfs-common -y
```

Create a folder

```
mkdir /nfs1
```

Temporary mounting of a nfs share :

```
mount -t nfs 192.168.19.129:/share1 /nfs1
```

Permanent mounting :

Open /etc/fstab file

And enter details

```
nano /etc/fstab
```

```
192.168.19.129:/share1  /nfs2   nfs     defaults        0 0
```

Save and exit

```
root@node1:~# systemctl daemon-reload
root@node1:~#
```

To check the mount :

```
root@node1:~# df -Th | grep nfs
192.168.19.129:/share1 nfs4    36G  1.3G   34G   4% /nfs2
```

Client OS : Rhel-8

Install package :

```
[root@node1 ~]# yum install nfs-utils -y
```

Check the share :

```
[root@node1 ~]# showmount -e 192.168.19.129
Export list for 192.168.19.129:
/share1 *
```

Mounting

Temp :

```
[root@node1 ~]# mkdir /nfs4
[root@node1 ~]#
[root@node1 ~]# mount -t nfs 192.168.19.129:/share1 /nfs4
```

Perm : fstab entry

```
192.168.19.129:/share1  /dir1   nfs     defaults        0 0
```

```
[root@node1 ~]# systemctl daemon-reload
```

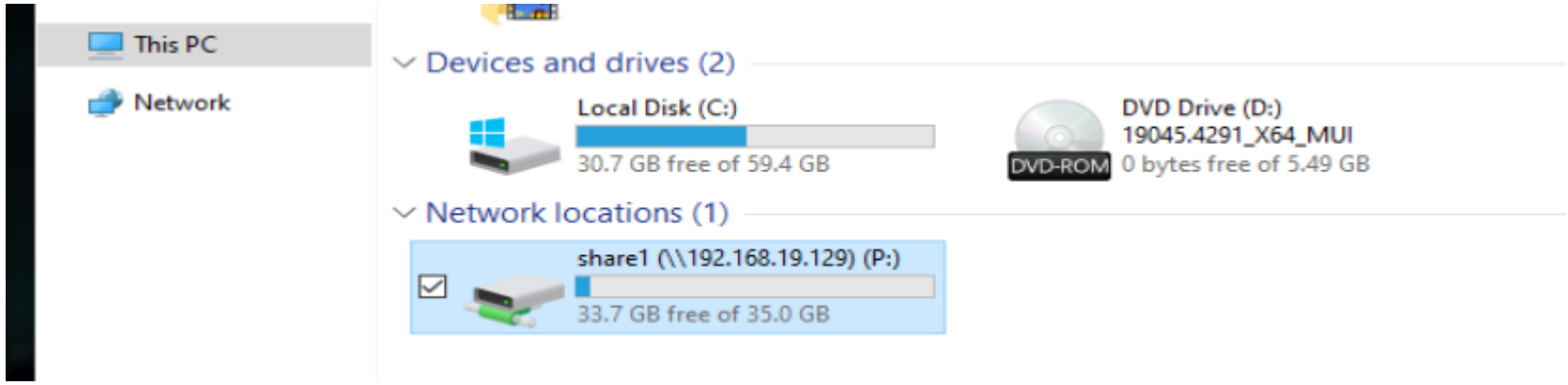
Client os : windows 10

Install nfs application : open control panel – go to windows feature and select services for nfs

Run command : appwiz.cpl

Connecting the nfs share :

```
C:\Users\Administrator>mount -o anon \\192.168.19.129\share1 P:
```


This PC
Network
Devices and drives (2)
Local Disk (C:)
30.7 GB free of 59.4 GB
DVD Drive (D:)
19045.4291_X64_MUI
DVD-ROM
0 bytes free of 5.49 GB
Network locations (1)
share1 (\\192.168.19.129) (P:)
33.7 GB free of 35.0 GB